Asteroseismology: Science Vision for stellar structure

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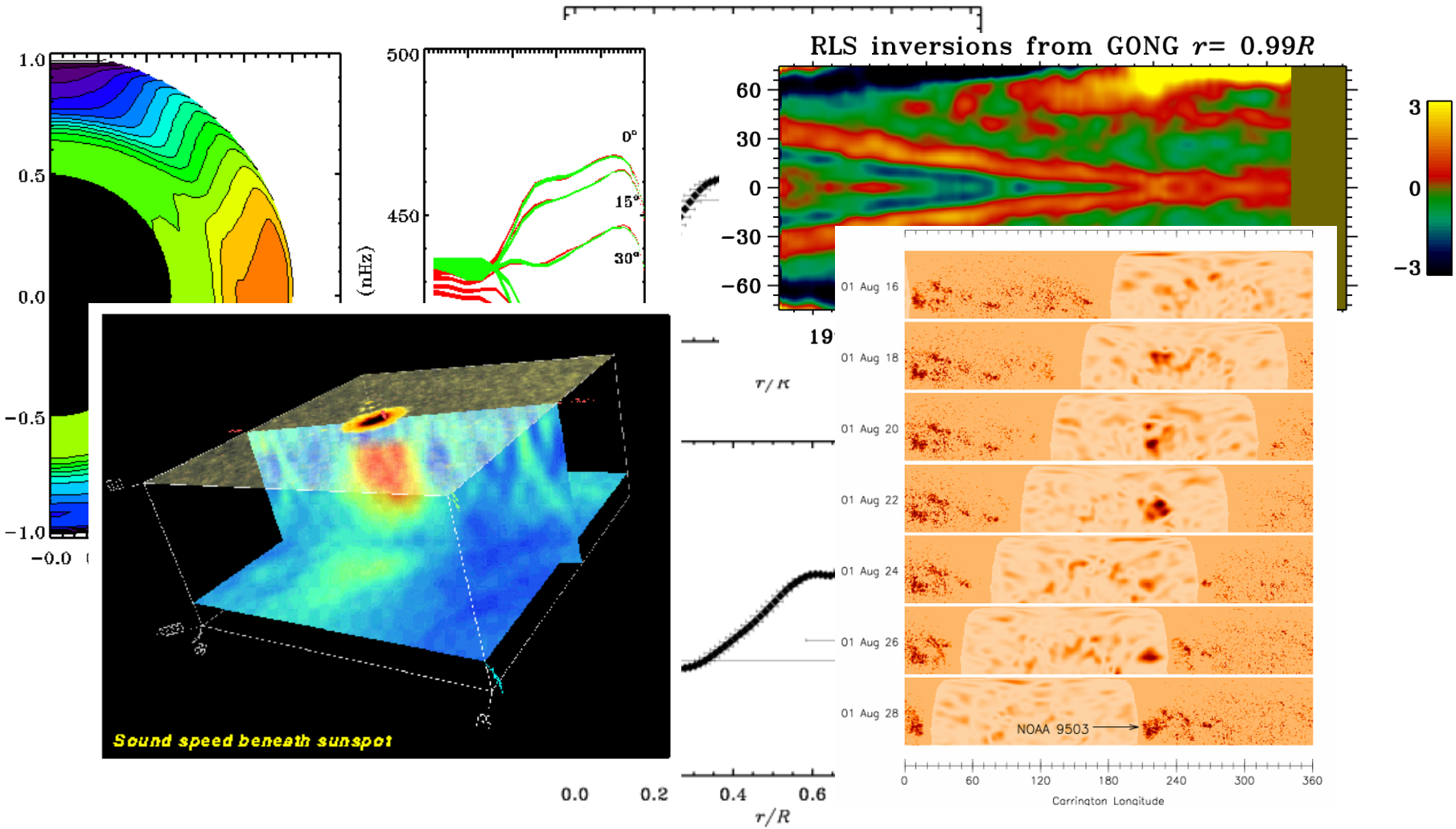
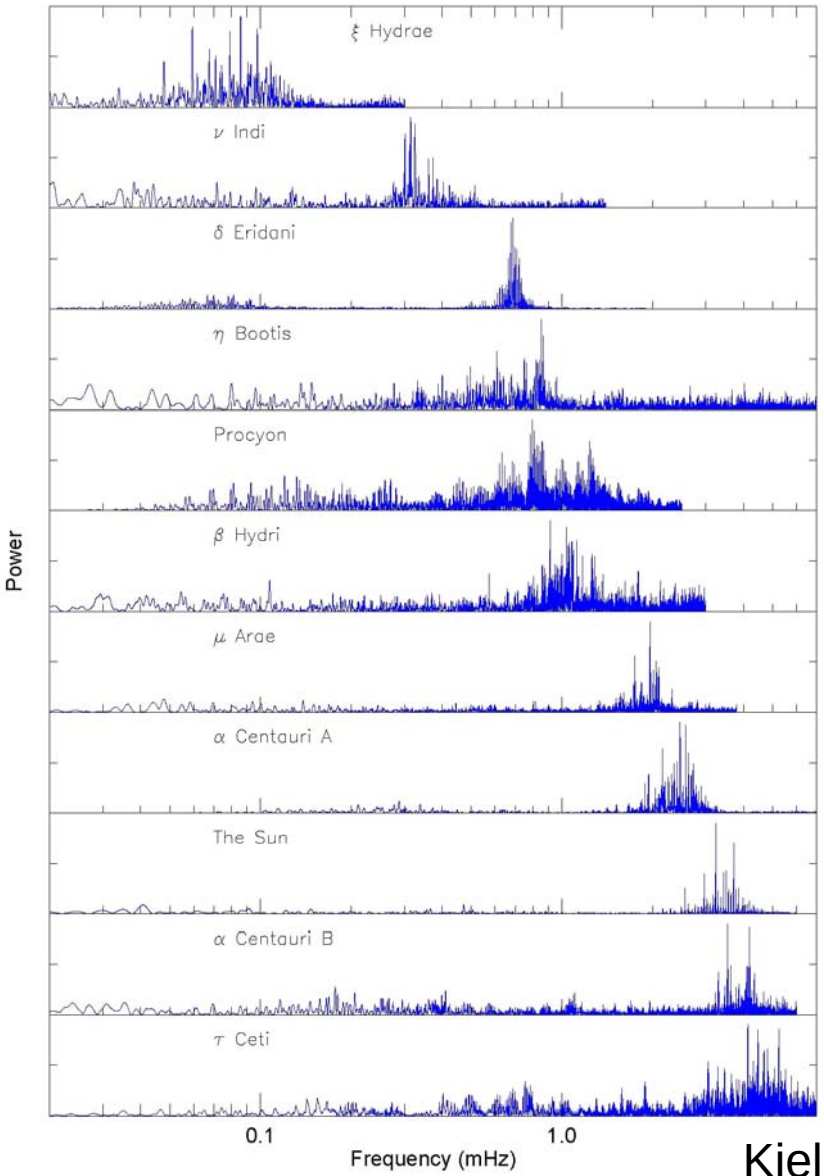
Helioseismology is Revolution !

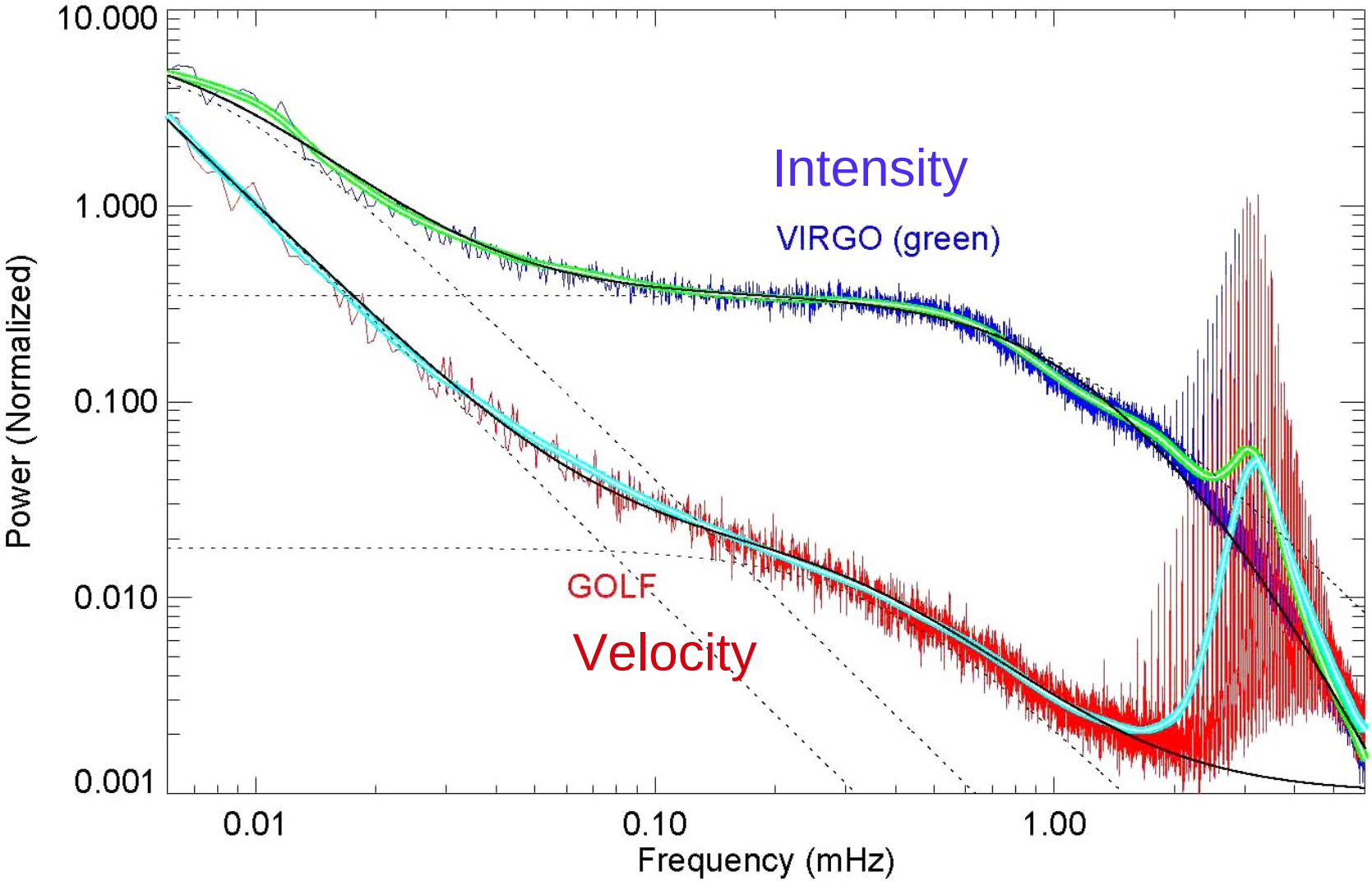


Stars do oscillate!

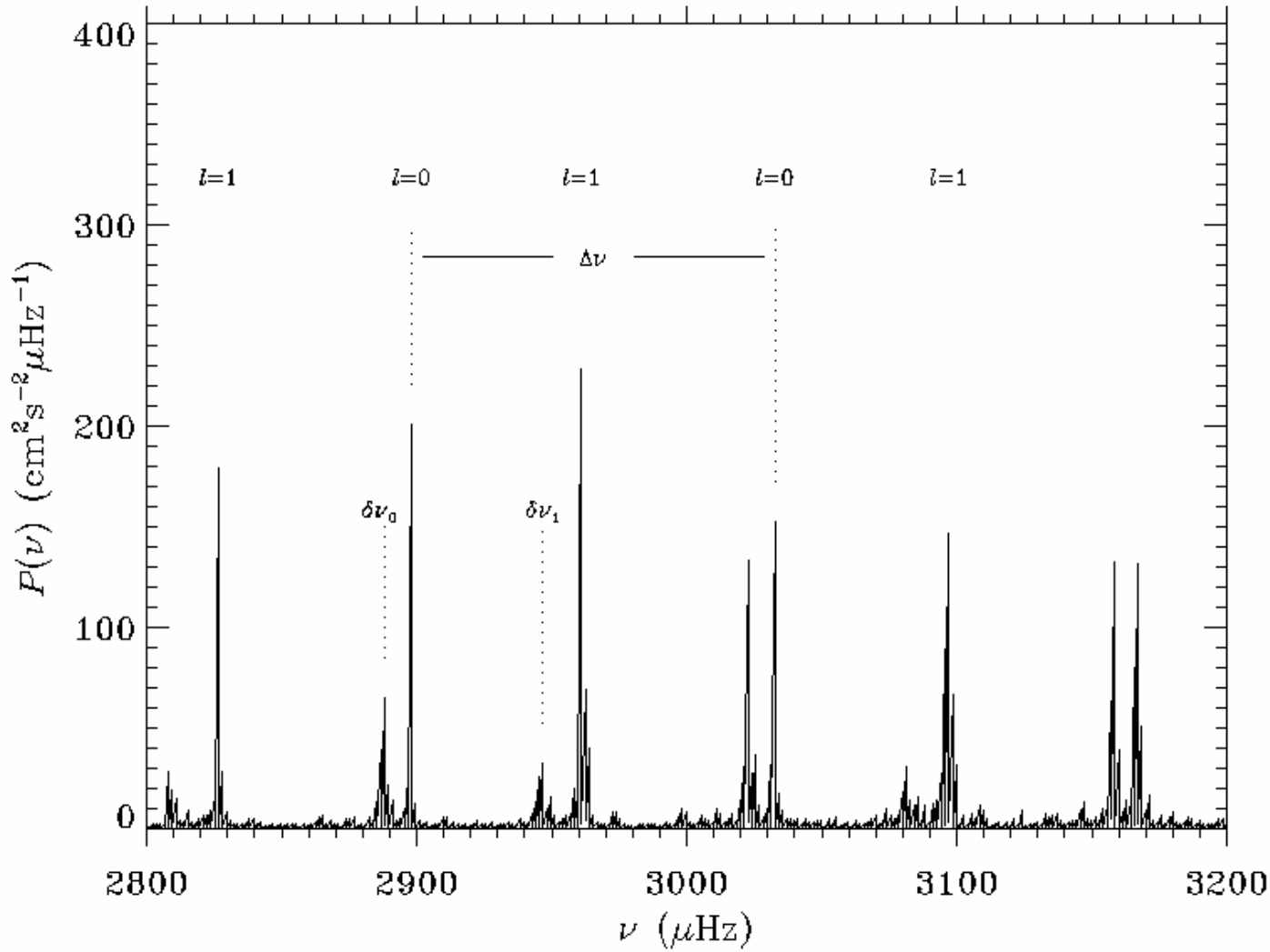


Kjeldsen & Bedding (2005)

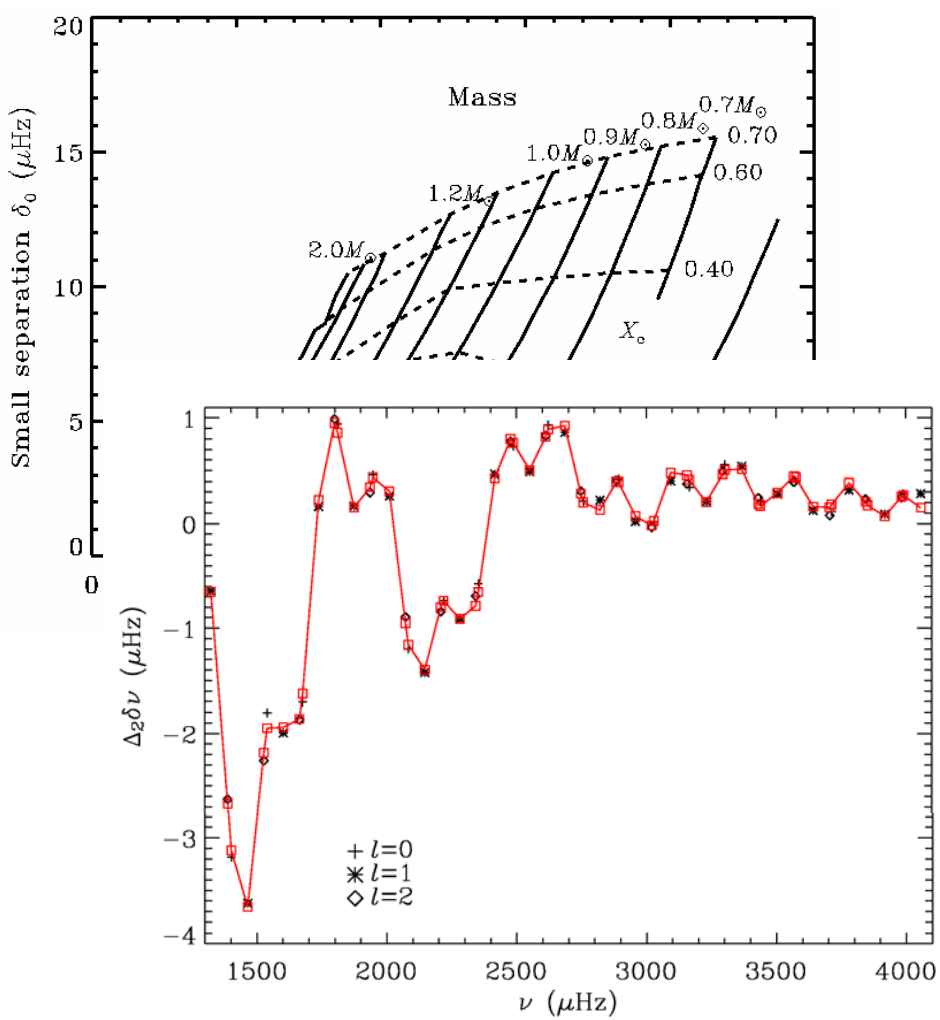
Helioseismology with SOHO

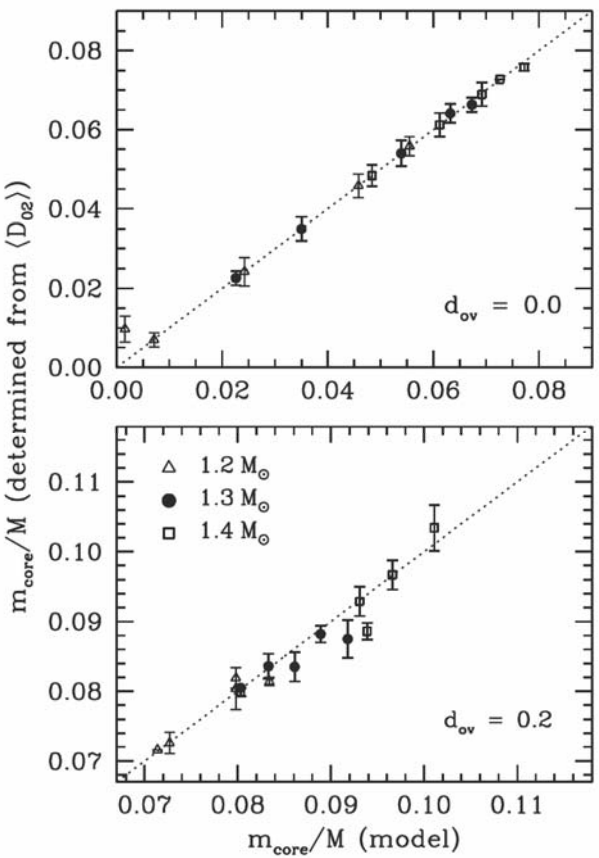


Seismic diagnostics



Asteroseismology is Revolution !

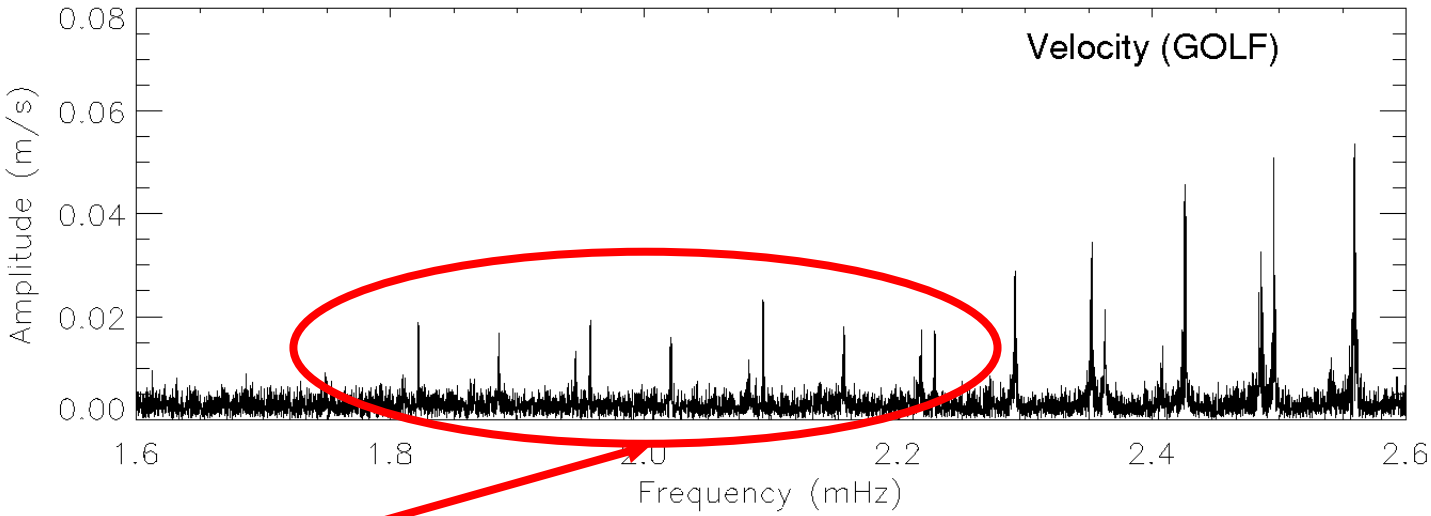




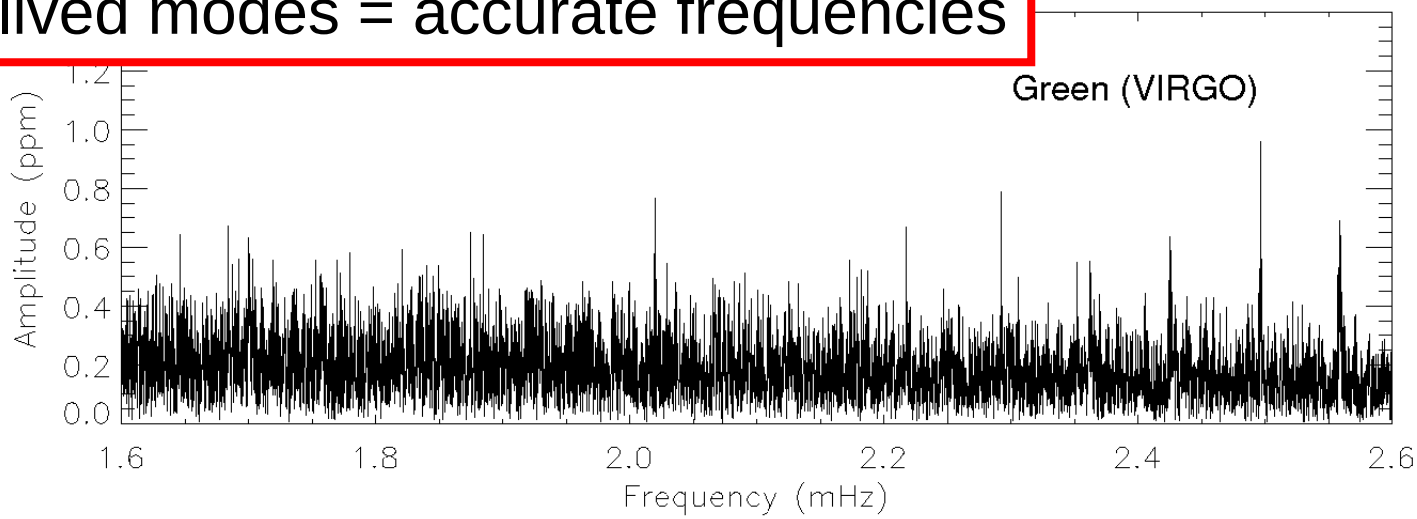
Scientific objectives

- Age determination: binaries, evolved stars
- Composition, etc… : clusters (age common)
- Stellar radii (impact for exoplanet radii)
- Diagnostic of convective cores
- Depth of convection and of second Helium ionization zones
- Rotation and internal structure
- Excitation mechanisms (convection)
- If images: tomography of differential rotation (interferometry)

Intensity vs Velocity: the solar case

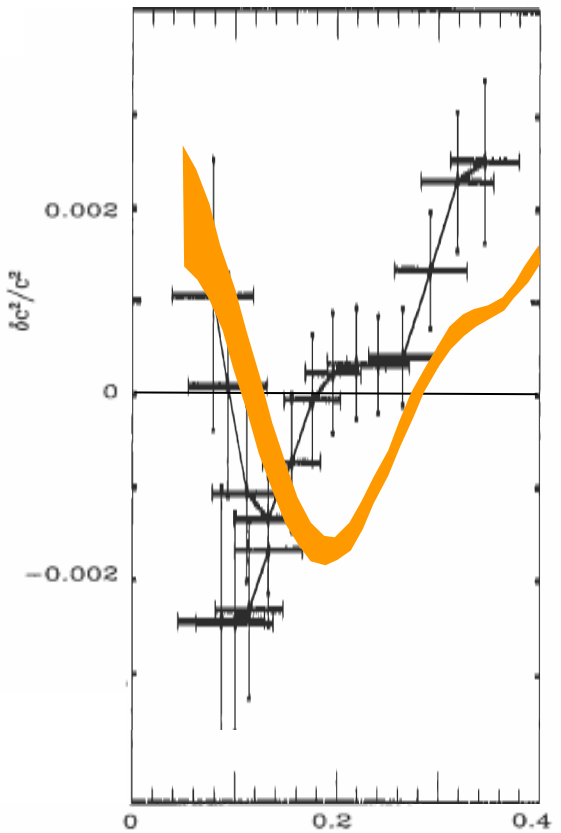


Long-lived modes = accurate frequencies

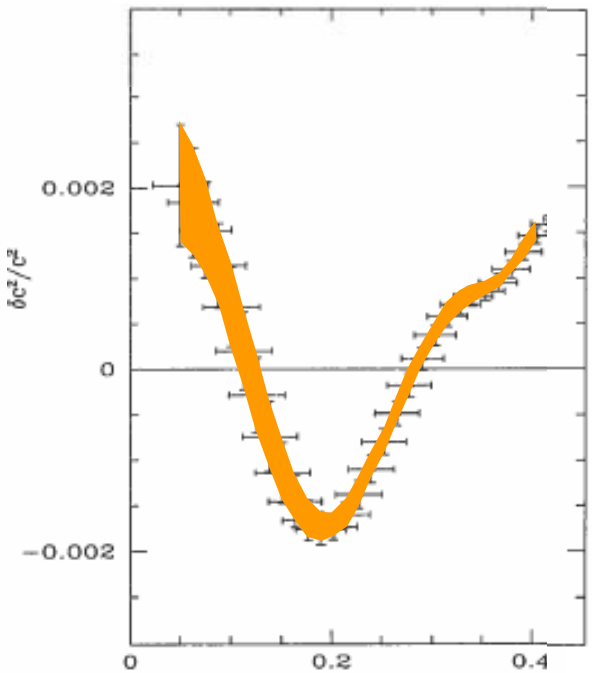


The impact on inversions

VIRGO / Intensity



GOLF / Velocity



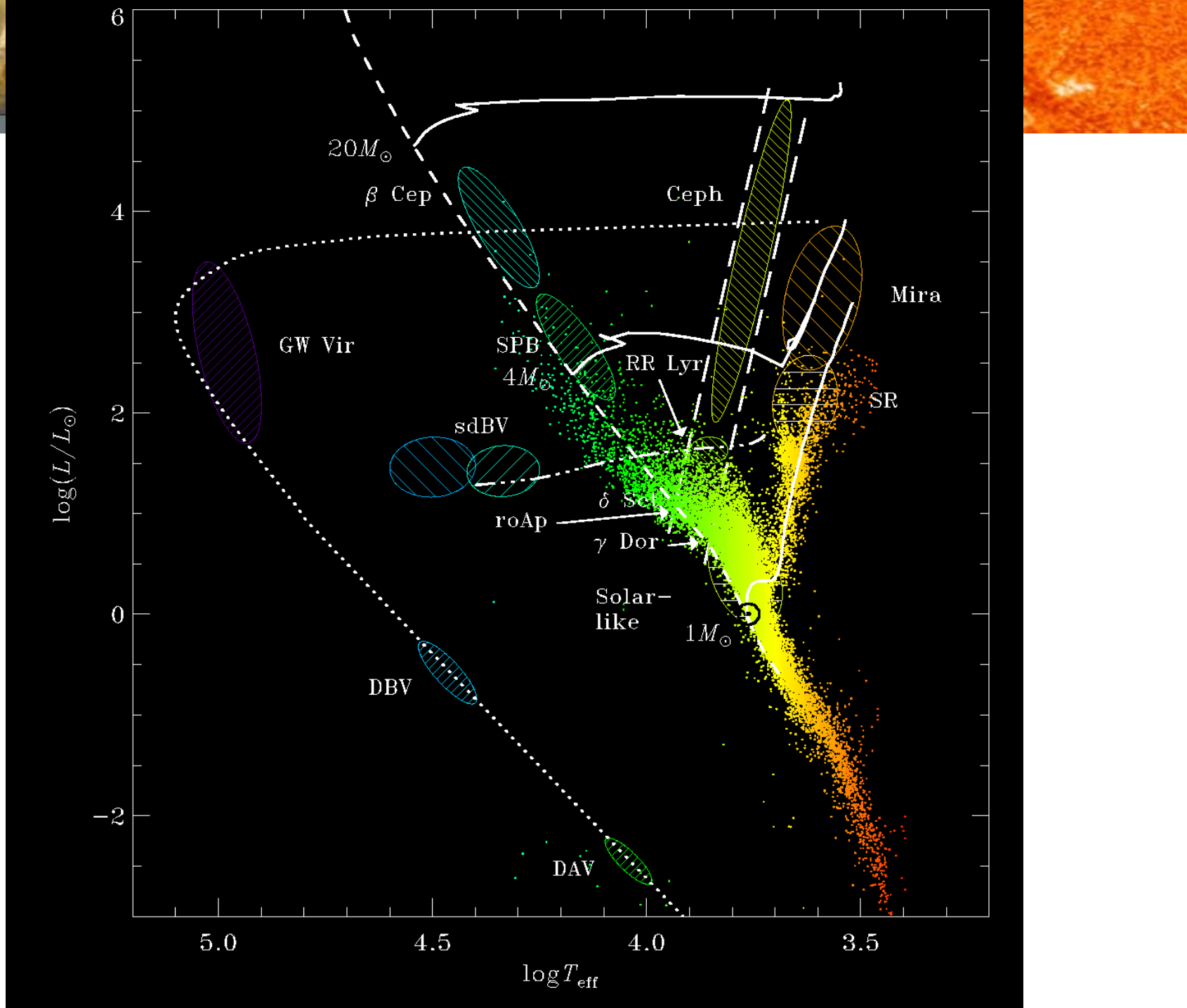
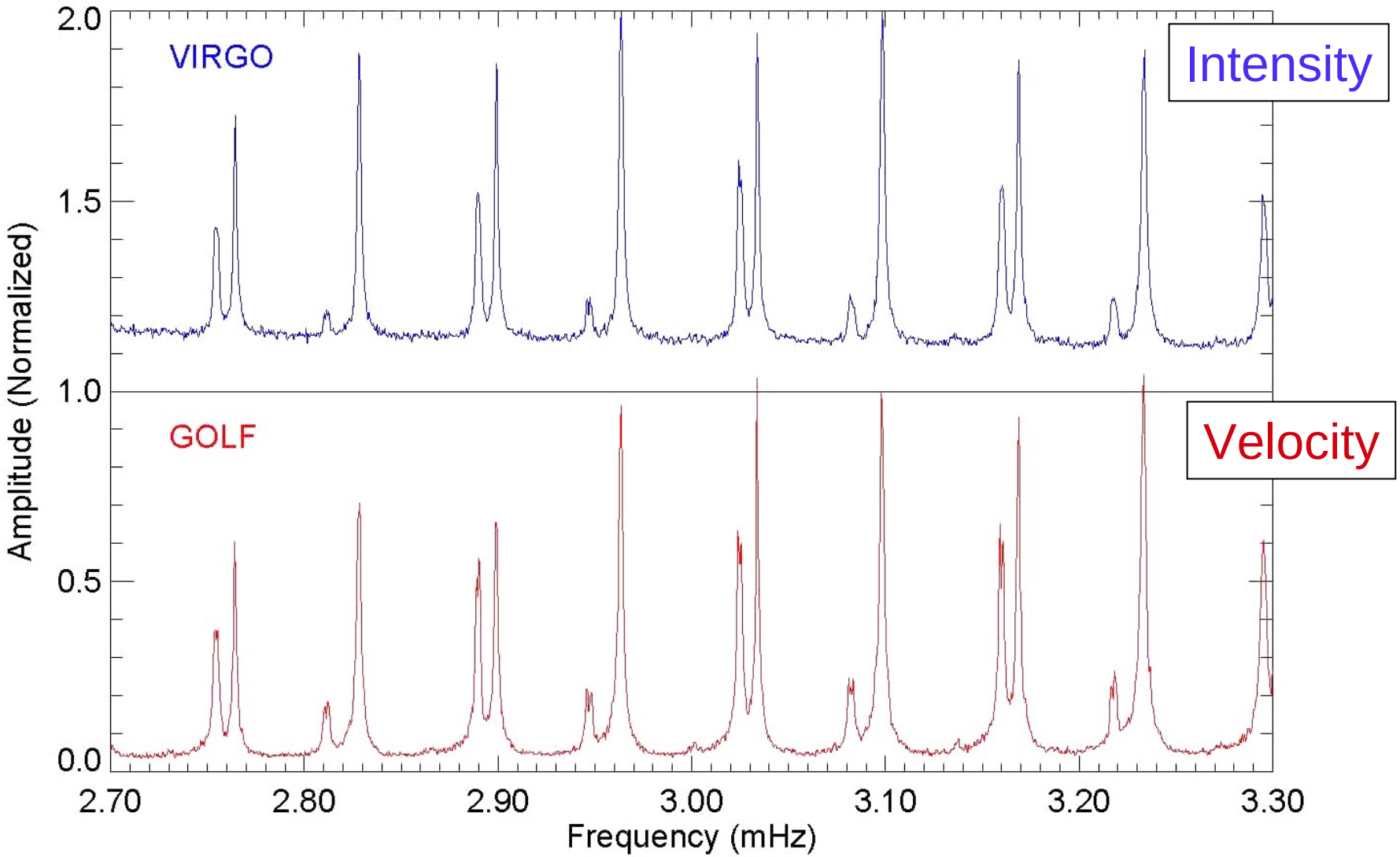
Adapted from Appourchaux et al and Gabriel et al (1996)

What do we need?

- Long observations (> 3 months to years)
- Spectroscopy (stellar radial velocities)
- Good sky coverage (network, Antartica)
- Limited day-night interruptions (space, Antartica)
- Stars with good parallaxes, good characteristics determination (Hipparcos and GAIA stars, binaries, open clusters)
- Images of stellar surfaces

Asteroseismology: a road map

- Intensity observations:
  - Multi-objects (CoRoT)
  - Cluster, multi-objects, faint objects (Kepler, PLATO)
- Stellar radial velocities:
  - Bright objects (Dedicated network, single site instrument in Antartica)
- Stellar images:
  - Interferometry leading to precise inversion for stellar structure and dynamics, rewarding like the Sun!
- Supported by GAIA !


6
4
2
0
−2
5.0
4.5
4.0
3.5
$\log T_{\rm eff}$
$\log(L/L_\odot)$
$20M_\odot$
β Cep
Ceph
Mira
GW Vir
SPB
$4M_\odot$
RR Lyr
SR
sdBV
δ Sct
roAp
γ Dor
Solar-
like
$1M_\odot$
DBV
DAV

Intensity vs Velocity: the solar case